

NIGHTINGALE COLLEGE

SCI 225 — Pathophysiology

Comprehensive Student Study Guide

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*This guide is designed to build deep conceptual understanding of disease processes.
Master the WHY behind each condition — mechanisms, adaptations, and clinical implications.*

How to Use This Guide

This study guide is organized by body system. Each section explains the underlying disease mechanisms, key clinical features, important distinctions between related conditions, and real-world analogies to anchor your understanding. Rather than memorizing isolated facts, your goal is to understand why a disease produces its signs and symptoms. That understanding will serve you in any clinical setting.

For each condition, ask yourself: What is broken? Why does that cause these symptoms? What would happen if left untreated? How is this condition distinguished from a similar one? These are the questions that drive deep learning in pathophysiology.

UNIT 1: HEMATOLOGY — DISORDERS OF BLOOD

1.1 Understanding Blood: The Basics

Blood is a connective tissue composed of cellular elements (red blood cells, white blood cells, and platelets) suspended in plasma. Its functions include oxygen transport, immune defense, hemostasis, and maintaining pH and fluid balance. Disruptions to any of these components produce the hematologic disorders covered in this unit.

Think of the bone marrow as a factory. Red blood cells (erythrocytes) are the delivery trucks for oxygen, white blood cells (leukocytes) are the security team, and platelets are the repair crew. When the factory malfunctions, production slows, the wrong products are made, or products are destroyed faster than they can be replaced.

1.2 Polycythemia Vera (PV)

Polycythemia vera is a myeloproliferative disorder characterized by excessive production of red blood cells, and to a lesser extent white blood cells and platelets, independent of normal regulatory signals. Unlike secondary polycythemia (which occurs as a compensatory response to hypoxia, such as at high altitude), PV is caused by a clonal mutation — most commonly in the JAK2 gene — that drives uncontrolled proliferation of hematopoietic stem cells.

Pathophysiology: The excess red blood cells dramatically increase blood viscosity. Think of normal blood as water flowing through a pipe, and PV blood as thick syrup. The sluggish, viscous flow predisposes patients to clot formation throughout the vascular system.

Clinical Manifestations: Patients typically present with ruddy (red) skin, hypertension, headaches, dizziness, pruritus after bathing (a hallmark symptom due to histamine release from basophils), and visual disturbances. Splenomegaly occurs as the spleen compensates for erythrocyte overproduction.

Complications: The most feared and most common cause of death in untreated PV is thrombosis — particularly cerebral thrombosis (stroke), which results from clot formation in the cerebral vasculature due to hyperviscosity. PV can also transform over time into myelofibrosis or acute myeloid leukemia (AML), representing disease progression.

Clinical Pearl

The classic triad of PV: elevated RBC mass + splenomegaly + normal oxygen saturation (rules out hypoxia-driven polycythemia). After hot baths, histamine release causes intense itching. No other common anemia or blood disorder produces this pattern.

1.3 Anemias — Overview and Classification

Anemia is defined as a reduction in the oxygen-carrying capacity of the blood, typically due to a decrease in red blood cell count, hemoglobin concentration, or hematocrit. It is not a diagnosis in itself but rather a sign of an underlying process. Classification is critical because treatment depends entirely on the cause.

Category	Mechanism	Example Conditions
Decreased Production	Bone marrow fails to make enough RBCs	Iron deficiency, B12 deficiency, aplastic anemia
Increased Destruction	RBCs are destroyed faster than replaced	Hemolytic anemia, sickle cell, thalassemia

Blood Loss	Acute or chronic hemorrhage	GI bleeding, heavy menses
Ineffective Erythropoiesis	RBCs produced but abnormal	Sideroblastic anemia, myelodysplasia

Iron Deficiency Anemia

The most common anemia worldwide. Iron is essential for hemoglobin synthesis; without adequate iron, red blood cells are produced in smaller sizes with less hemoglobin content. This produces a microcytic, hypochromic (small, pale) anemia.

Common Causes: In infants and toddlers, excessive cow's milk consumption is a classic cause. Cow's milk is low in iron, inhibits iron absorption, and can cause occult GI blood loss. Children aged 6 months to 2 years are at highest risk. In menstruating women, excessive blood loss is common. In older adults, GI malignancy must be ruled out.

Laboratory Findings: Serum ferritin is the most direct measure of iron stores — it reflects the body's iron reserves in storage form. A low ferritin is virtually diagnostic of iron deficiency. Transferrin saturation (iron/TIBC ratio) and total iron-binding capacity (TIBC, which rises in deficiency) are supporting tests. A bone marrow biopsy with Prussian blue staining directly visualizes iron stores but is rarely needed clinically.

Analogy

Think of ferritin as the pantry stockpile and transferrin as the delivery truck. In iron deficiency, the pantry is empty (low ferritin) and more trucks are ordered to seek iron (high TIBC). Transferrin saturation tells you how full the trucks are — in deficiency, trucks are mostly empty.

Pernicious Anemia (Vitamin B12 Deficiency)

Vitamin B12 is required for DNA synthesis; its absence impairs the rapid cell division needed to produce normal red blood cells. The result is large, immature cells (megaloblasts) that are produced in reduced numbers — a macrocytic, normochromic anemia.

Pathophysiology of Pernicious Anemia Specifically: Pernicious anemia is an autoimmune condition in which the stomach's parietal cells are destroyed, eliminating production of intrinsic factor. Intrinsic factor is the glycoprotein produced by gastric parietal cells that binds dietary B12 and escorts it to the terminal ileum for absorption. Without intrinsic factor, dietary B12 cannot be absorbed regardless of how much is consumed.

Clinical Features: Beyond the anemia itself, B12 deficiency causes neurological damage (subacute combined degeneration of the spinal cord), a beefy red, smooth tongue (glossitis), and lemon-yellow skin (mild jaundice from ineffective erythropoiesis). Neurological symptoms are irreversible if untreated, and their progressive severity can ultimately be fatal. The macrocytic RBCs in B12 and folate deficiency are due to defective DNA synthesis, not abnormal hemoglobin synthesis.

Sideroblastic Anemia

In sideroblastic anemia, iron is taken up by developing red blood cells but cannot be incorporated into hemoglobin properly. Iron accumulates in the mitochondria of erythroid precursors, forming ring sideroblasts visible on bone marrow examination. The result is ineffective iron uptake into hemoglobin combined with abnormal hemoglobin production, even though iron stores are adequate or elevated. Causes include alcohol use, lead poisoning, isoniazid therapy, and genetic mutations.

Hemolytic Anemia

Hemolytic anemia results from premature destruction of red blood cells. When RBC lifespan drops below the normal 120 days, the bone marrow attempts to compensate with increased erythropoiesis (evidenced by reticulocytosis), but cannot keep up. Causes include autoimmune attack (warm/cold antibody), mechanical destruction, enzyme deficiencies (G6PD), and membrane defects (hereditary

spherocytosis). In normocytic-normochromic anemia — cells of normal size and color — hemolytic anemia is a key consideration.

Aplastic Anemia

Aplastic anemia is a failure of the bone marrow to produce all blood cell lines (pancytopenia). The marrow becomes hypocellular, replaced by fat. Causes include autoimmune destruction of hematopoietic stem cells, radiation, toxins, and idiopathic mechanisms. Because all cell lines are affected, patients experience anemia, infections (from neutropenia), and bleeding (from thrombocytopenia) simultaneously.

1.4 Thalassemias

Thalassemias are a group of inherited disorders characterized by defective synthesis of one or more hemoglobin chains. They are inherited in an autosomal recessive pattern. Normal adult hemoglobin (HbA) consists of two alpha and two beta chains. Mutations that reduce or eliminate the production of one chain type lead to an imbalance, with excess unpaired chains precipitating and damaging red blood cells.

Alpha Thalassemia	Beta Thalassemia
Defect in alpha chain production	Defect in beta chain production
4 alpha genes total (2 per chromosome)	2 beta genes total (1 per chromosome)
Severity depends on number of genes deleted	Minor (1 gene) or Major (2 genes affected)
Alpha thalassemia major (all 4 genes deleted) = hydrops fetalis — incompatible with life	Beta major (Cooley anemia) = severe transfusion-dependent anemia
Hemoglobin H disease (3 genes deleted) = moderate anemia	Beta minor (trait) = mild, often asymptomatic

Clinical Key

Hydrops fetalis — massive edema, cardiomegaly, hepatomegaly, and intrauterine heart failure — occurs exclusively in alpha thalassemia major because all four alpha genes are deleted, making it impossible to produce any functional hemoglobin. It is universally fatal in utero or shortly after birth.

1.5 Sickle Cell Disease

Sickle cell disease is caused by a single point mutation in the beta-globin gene (glutamic acid replaced by valine), producing hemoglobin S (HbS). When HbS is deoxygenated, it polymerizes and distorts the red blood cell into a rigid, sickle shape. These sickled cells are fragile, have a shortened lifespan (~20 days instead of 120), and obstruct small blood vessels.

Sickle cell anemia (HbSS) occurs when a person inherits two HbS alleles. The trait (HbAS) occurs when one HbS and one HbA allele are inherited; the mother in the genetic transmission question has HbS and HbA — she carries the trait, meaning the child could have inherited two HbS alleles if the father also contributed one.

Types of Sickle Cell Crises:

- Vaso-occlusive (pain crisis): The most common. Sickled cells block microvascular flow, causing severe ischemic pain. This is the hallmark symptom of crisis.

- **Aplastic crisis:** Triggered by parvovirus B19 infection, which temporarily halts erythropoiesis. Dangerous because sickled RBCs are already short-lived.
- **Sequestration crisis:** Occurs mainly in young children. The spleen suddenly traps large volumes of sickled cells, causing acute splenic enlargement and circulatory collapse.
- **Hyperhemolytic crisis:** Accelerated hemolysis leads to rapidly falling hemoglobin.

Sickled cells are removed from circulation primarily by the spleen, which explains the progressive functional asplenia seen in older patients with sickle cell disease. The spleen is damaged over time by repeated infarctions.

Population Note

Sickle cell disease disproportionately affects individuals of Black/African descent. In the United States, universal newborn screening should specifically prioritize this population.

1.6 Leukemia

Leukemia is a malignant clonal proliferation of white blood cell precursors in the bone marrow. The uncontrolled growth of immature blasts crowds out normal hematopoietic cells, causing anemia, thrombocytopenia, and immunosuppression. The four main subtypes are defined by cell lineage (myeloid vs lymphoid) and disease course (acute vs chronic).

Type	Common Age	Key Feature
ALL (Acute Lymphoblastic)	Children (peak 2–5 yrs)	Most common childhood leukemia; associated with Down syndrome
AML (Acute Myeloid)	Adults; PV can transform to AML	Can arise from myeloproliferative disorders
CML (Chronic Myeloid)	Middle-aged adults	Philadelphia chromosome (BCR-ABL translocation)
CLL (Chronic Lymphocytic)	Elderly	Most common adult leukemia; slow progression

Clinical Manifestations of Leukemia: Fatigue and pallor (anemia from marrow crowding), petechiae and bleeding (thrombocytopenia), fever and recurrent infections (loss of functional neutrophils), bone pain (marrow expansion), and lymphadenopathy/hepatosplenomegaly. In children with ALL, jaundice may also occur.

Association

Down syndrome (trisomy 21) carries a significantly elevated risk of leukemia, particularly ALL and AML. When a child presents with leukemia, the nurse should assess for signs of Down syndrome.

1.7 Lymphoma

Lymphomas are malignancies of the lymphatic system, broadly divided into Hodgkin and non-Hodgkin lymphoma. They differ in pathology, epidemiology, and treatment.

Hodgkin Lymphoma (HL): Defined pathologically by the presence of Reed-Sternberg cells — large, binucleated cells that look like 'owl eyes' under the microscope. HL presents classically with painless cervical (neck) lymphadenopathy, fever, night sweats, and unintentional weight loss (the 'B symptoms'). It often involves a mediastinal mass. HL is highly curable with modern therapy.

Burkitt Lymphoma: An aggressive non-Hodgkin lymphoma associated with Epstein-Barr virus. It is characterized by an extremely rapid doubling time and often involves the jaw (endemic form in Africa) or abdomen.

1.8 Multiple Myeloma

Multiple myeloma is a malignant proliferation of plasma cells in the bone marrow. These abnormal plasma cells produce large amounts of non-functional immunoglobulin (paraprotein or M-protein). The bone destruction caused by myeloma cells releasing osteoclast-activating factors is the primary source of the severe bone pain characteristic of the disease. This bone destruction also leads to hypercalcemia, pathological fractures, and anemia.

1.9 Thrombocytopenia and Immune Thrombocytopenic Purpura (ITP)

Thrombocytopenia is defined as a platelet count below 100,000 per cubic millimeter (some sources use 150,000). Platelets are essential for primary hemostasis — forming the initial platelet plug at sites of vessel injury.

Immune Thrombocytopenic Purpura (ITP) is an autoimmune condition in which antibodies (typically IgG) coat platelets and mark them for destruction by the spleen. In children, it often follows a viral infection and resolves spontaneously. In adults, it tends to be chronic. Clinical features include petechiae, ecchymoses (bruising), and mucosal bleeding. The most serious complication is intracranial bleeding, which can be life-threatening even though it is rare.

Clinical Presentation

A child with ITP classically presents with bruising on the legs and trunk, petechiae, and nosebleeds following a recent viral illness. The platelets are being destroyed by antibodies — they are not being produced incorrectly. This distinguishes ITP from aplastic anemia (where production itself fails).

1.10 Hemophilia

Hemophilia is an X-linked recessive bleeding disorder caused by deficiency of a specific clotting factor. Because the gene is on the X chromosome, males (who have only one X) are primarily affected, while females are typically carriers.

Hemophilia A	Hemophilia B
Deficiency of Factor VIII	Deficiency of Factor IX
'Classic hemophilia'	'Christmas disease' (named after a patient)
More common (85% of cases)	Less common

1.11 Hemolytic Disease of the Newborn (HDN) and Rh Incompatibility

HDN occurs when maternal antibodies cross the placenta and destroy fetal red blood cells. The most clinically significant scenario is Rh incompatibility: a Rh-negative mother carries a Rh-positive fetus. During delivery (or miscarriage), fetal blood can enter the maternal circulation. The mother's immune system recognizes the Rh antigen as foreign and produces anti-D antibodies. In subsequent Rh-positive pregnancies, these antibodies cross the placenta and cause fetal red blood cell destruction.

The Coombs test (direct and indirect) identifies antibodies attached to red blood cells and is the key laboratory test used to detect infants at risk for HDN.

Note: The mother being Rh-positive and fetus Rh-negative carries no risk — the mother already has the antigen and produces no antibodies against it. Risk arises only when the MOTHER lacks an antigen that the FETUS has.

1.12 Infectious Mononucleosis

Caused by Epstein-Barr virus (EBV), infectious mononucleosis — 'mono' or the 'kissing disease' — presents in adolescents and young adults with fever, pharyngitis (sore throat), lymphadenopathy (especially posterior cervical nodes), profound fatigue, and hepatosplenomegaly. A morbilliform (maculopapular) rash may appear, particularly if the patient is mistakenly treated with ampicillin.

Transmission is through oral secretions. Prevention focuses on not sharing drinking glasses or eating utensils. Splenomegaly poses a risk for splenic rupture with contact sports, which must be avoided.

1.13 Leukocyte Disorders

Leukocytosis: An elevated white blood cell count (above 11,000/mm³), typically representing an immune response to infection, inflammation, or malignancy.

Eosinophilia: An elevated eosinophil count, most commonly associated with parasitic infections (helminths) and allergic reactions (hypersensitivity). Eosinophils are mediators of allergic inflammation and are recruited to combat parasites.

UNIT 2: CARDIOVASCULAR PATHOPHYSIOLOGY

2.1 Hypertension

Hypertension is defined as a sustained blood pressure above 130/80 mmHg. Normal adult blood pressure is a systolic below 120 mmHg and diastolic below 80 mmHg. Hypertension is classified as primary (essential) — no identifiable cause, accounting for the vast majority of adult cases — or secondary, where an underlying condition drives elevated pressure.

Primary (Essential) Hypertension: No single identifiable cause; multifactorial, involving genetic predisposition, sodium retention, sympathetic activation, and vascular remodeling. It is by far the most common form.

Secondary Hypertension: Hypertension driven by an identifiable cause. In children, secondary hypertension is far more common than in adults, with renal disease being the leading culprit. When a child is found to have persistent hypertension, the workup should focus primarily on the kidneys.

Pulse Pressure and Valve Disorders: Blood pressure findings can indicate valve disorders. A widened pulse pressure (systolic and diastolic far apart) combined with bounding, throbbing peripheral pulses suggests aortic regurgitation — blood leaking back into the left ventricle causes the ventricle to eject a larger-than-normal volume, raising systolic pressure, while the backflow lowers diastolic pressure.

2.2 Coronary Artery Disease and Angina

Coronary artery disease (CAD) results from atherosclerosis — the buildup of lipid-laden plaques within the walls of coronary arteries. These plaques narrow the lumen and reduce blood flow to the myocardium. Modifiable risk factors include smoking, hyperlipidemia, hypertension, diabetes, and obesity. Non-modifiable factors include age, sex (male), and family history.

Angina Pectoris: Chest pain that occurs when myocardial oxygen demand exceeds supply. The classic presentation is substernal chest pain that may radiate to the left shoulder, arm, or jaw.

- **Stable angina:** Predictable, reproducible pain triggered by exertion, relieved by rest. The coronary artery is narrowed but the plaque is stable.
- **Unstable angina:** Occurs at rest or with minimal exertion; increasingly frequent. Represents a rupturing or vulnerable plaque with dynamic changes in flow. A medical emergency.
- **Prinzmetal (variant) angina:** Caused by coronary artery spasm, often occurring at rest and at night. Associated with preserved coronary anatomy.

2.3 Myocardial Infarction (MI)

MI results from complete occlusion of a coronary artery, most commonly due to rupture of an atherosclerotic plaque followed by platelet aggregation and thrombus formation. The zone of myocardium supplied by the blocked artery undergoes ischemia, injury, and ultimately necrosis if blood flow is not restored.

Autopsy Finding

At autopsy, MI most commonly reveals platelet aggregation within an atherosclerotic coronary artery — not emboli from the aorta, which is rare. The pathological cascade is plaque rupture → platelet adhesion → thrombus → occlusion.

Shock following MI is primarily cardiogenic — the damaged myocardium can no longer pump effectively, reducing cardiac output. This is distinct from decreased heart rate (which is a compensatory response, not the primary mechanism), peripheral vasodilation (more characteristic of septic or neurogenic shock), or capillary leak (anaphylactic shock).

2.4 Heart Failure

Heart failure occurs when the heart cannot pump sufficient blood to meet the body's metabolic demands, or can only do so at elevated filling pressures. It is classified as left-sided or right-sided based on which ventricle is primarily affected.

Left Heart Failure	Right Heart Failure
Left ventricle cannot eject blood forward effectively	Right ventricle cannot push blood through the lungs
Blood backs up into the pulmonary vasculature	Blood backs up into the systemic venous circulation
Pulmonary vascular congestion → pulmonary edema	Peripheral edema, jugular venous distension, hepatomegaly
Symptoms: dyspnea, orthopnea, cough, frothy sputum	Symptoms: dependent edema, ascites, distended neck veins
Cause: MI, aortic stenosis, hypertension	Often caused by left heart failure or pulmonary HTN (cor pulmonale)

Infants and Children with Heart Failure: In a 1-year-old with a ventricular septal defect developing left-sided heart failure, the primary symptom is failure to thrive — the infant cannot feed effectively because increased respiratory effort from pulmonary congestion exhausts the baby's energy reserves before adequate nutrition is obtained.

High-Output vs Low-Output Heart Failure: In a patient with chronic pulmonary disease and elevated pulmonary vascular resistance, the right heart must work against increased afterload, leading to right-sided heart failure (cor pulmonale) — a form of low-output failure.

2.5 Shock

Shock is a state of circulatory failure resulting in inadequate oxygen delivery to tissues. All forms of shock, regardless of cause, share the endpoint of cellular hypoxia. Understanding the mechanism helps distinguish the types.

Type	Primary Mechanism	Classic Scenario
Hypovolemic	Loss of blood volume	Hemorrhage, burns, dehydration
Cardiogenic	Pump failure	MI, severe cardiomyopathy
Distributive/Septic	Vasodilation and maldistribution	Gram-negative bacteremia (endotoxin)
Neurogenic	Loss of sympathetic tone → vasodilation	C-5 spinal cord transection
Anaphylactic	Massive histamine release → vasodilation + capillary leak	Bee sting, penicillin allergy

Clinical Key

Neurogenic shock — as seen in cervical spinal cord injury — results from loss of sympathetic outflow below the level of injury, causing unopposed vasodilation and bradycardia. It is NOT hypovolemic despite low BP; the patient loses sympathetic tone, not blood volume. Septic shock also causes vasodilation, but

through bacterial toxins and cytokines, not loss of neural control.

2.6 Valvular Heart Disease

Valvular disorders involve either stenosis (narrowing, obstructing forward flow) or regurgitation (incompetence, allowing backward flow). The most common cause of combined stenosis and regurgitation is rheumatic fever/rheumatic heart disease, which causes scarring and fibrosis of the valve leaflets. The most common cardiac valve disease in the United States is mitral valve prolapse.

Valvular Problem	Hemodynamic Consequence
Mitral stenosis	Obstruction of flow from LA to LV; LA enlarges; pulmonary congestion
Mitral regurgitation	Blood flows backward into LA during systole; LV dilates
Aortic stenosis	LV must generate higher pressure to overcome outflow obstruction; LV hypertrophies
Aortic regurgitation	Blood leaks back into LV; widened pulse pressure; bounding pulses

In the case of a 72-year-old with hypertension, atherosclerosis, and backflow of blood into the left ventricle demonstrated by echocardiogram — this describes aortic regurgitation. The aortic valve is between the aorta and the left ventricle; backflow goes from aorta into the LV.

2.7 Congenital Heart Defects

Congenital heart defects (CHDs) arise from disruptions in normal cardiac development during fetal life. Intrauterine exposures that increase risk include viral infections (rubella), maternal diabetes, and certain drugs. The most common CHD is ventricular septal defect (VSD).

Ventricular Septal Defect (VSD)

A hole in the wall between the ventricles. Because left ventricular pressure normally exceeds right, blood shunts left-to-right. This increases pulmonary blood flow and can cause pulmonary hypertension over time. Infants present with poor feeding, fatigue, and failure to thrive. Down syndrome is a common genetic association with VSD.

Patent Ductus Arteriosus (PDA)

The ductus arteriosus is a vessel connecting the aorta to the pulmonary artery in fetal circulation, allowing blood to bypass the non-functional fetal lungs. It normally closes shortly after birth. If it remains patent (open), blood from the high-pressure aorta shunts left-to-right into the pulmonary artery, creating a left-to-right shunt and increasing pulmonary flow.

Tetralogy of Fallot (TOF)

A complex defect comprising four abnormalities: pulmonary stenosis, right ventricular hypertrophy, overriding aorta, and VSD. The pulmonary stenosis is the critical lesion — it obstructs blood flow to the lungs, causing right-to-left shunting through the VSD. Blood bypasses the lungs, delivering deoxygenated blood to the body. The cardinal features are cyanosis and hypoxia. 'Tet spells' occur when infants cry or exert themselves, acutely increasing right-to-left shunting.

Tricuspid Atresia

Failure of the tricuspid valve to develop, meaning no blood can flow from the right atrium to the right ventricle. Blood must shunt through an atrial septal defect to reach the left heart and pulmonary circulation. Presents with cyanosis from birth.

Pulmonic Stenosis

Narrowing of the pulmonary valve, obstructing right ventricular outflow. The 'valve' is the specific site of narrowing; other anatomical locations can also be stenotic (infundibular, supra-ventricular). Presents with murmur and possibly cyanosis depending on severity.

Coarctation of the Aorta

A narrowing of the aorta, most commonly just distal to the left subclavian artery. This creates a pressure differential: hypertension in the upper extremities and weak or absent femoral pulses in the lower extremities. This pressure difference is the clinical key to diagnosis.

Transposition of the Great Arteries

The aorta arises from the right ventricle and the pulmonary artery from the left ventricle — completely reversed. This creates two parallel, non-mixing circulatory loops. The right side receives deoxygenated blood from the body and pumps it back to the body; the left side receives oxygenated blood from the lungs and sends it back to the lungs. Survival depends entirely on mixing at a septal defect or PDA. The characteristic defect: the aorta leaves the right ventricle.

Atrioventricular Canal (AVC) Defect

Results from incomplete fusion of the endocardial cushions during fetal development, leaving holes between all four chambers and malformed mitral/tricuspid valves. Common in Down syndrome.

Truncus Arteriosus

A single common arterial trunk arises from both ventricles, receiving blood from both. When accompanied by a VSD (which is always the case), blood from both ventricles mixes in this common vessel before distribution to the body and lungs.

2.8 Fetal Circulation: Ductus Arteriosus vs Foramen Ovale

Two key structures maintain fetal circulation by bypassing the non-functioning fetal lungs. The foramen ovale is an opening between the right and left atria that allows blood to bypass the right ventricle. The ductus arteriosus connects the pulmonary artery to the aorta, shunting blood away from the lungs. Both normally close after birth as pulmonary vascular resistance drops with the first breath.

2.9 Kawasaki Disease

An acute febrile vasculitis of unknown cause affecting medium-sized vessels — specifically the small capillaries, arteries, and veins — primarily in children under 5 years old. It is the leading cause of acquired heart disease in children in developed countries. The most dangerous complication is coronary artery aneurysm formation. Classic features include prolonged fever, rash, conjunctivitis, cracked lips, strawberry tongue, and cervical lymphadenopathy.

2.10 Raynaud Disease

Raynaud disease is characterized by episodic, exaggerated vasospasm of small arteries and arterioles in the fingers and toes in response to cold or emotional stress. The classic color change progression is white (ischemia from vasospasm) → blue (cyanosis) → red (reperfusion hyperemia). Cold exposure is the primary trigger to avoid. Smoking exacerbates vasospasm and must be discouraged. Calcium channel blockers reduce vasospasm and are a treatment option.

2.11 Venous Disorders and Thromboembolism

Thrombus vs Embolus: A thrombus is a clot fixed to a vessel wall. An embolus is a detached, traveling clot (or other material). A thromboembolus is a thrombus that has broken free and embolized.

Pulmonary Embolism (PE): Most commonly results from a deep vein thrombus (DVT) in the lower extremities that breaks free and travels to the pulmonary vasculature. Prevention is centered on preventing DVT formation — through early mobilization, anticoagulation, and compression devices. Clinical features include sudden-onset chest pain, tachycardia, tachypnea, dyspnea, fever, and hemoptysis. In a surgical patient, preventing PE starts with preventing DVT.

Chronic Venous Insufficiency: Pooling of blood in the lower extremities due to incompetent venous valves. Leads to skin changes including hyperpigmentation (brown discoloration from hemosiderin deposition), chronic edema, and risk of ulceration.

Superior Vena Cava Syndrome (SVCS): Progressive occlusion of the superior vena cava, most commonly from a mediastinal tumor (especially lung cancer or lymphoma). Occlusion impairs venous return from the head, neck, and upper extremities, causing facial swelling, dilated neck veins, and upper extremity edema.

2.12 Infective Endocarditis

Infective endocarditis is infection of the endocardium, most commonly the cardiac valves. The most common causative organism is *Staphylococcus aureus*, though *Streptococcus viridans* is also important. Classic presentation: fever, new cardiac murmur, and petechial skin lesions (from septic emboli). The vast majority of cases are bacterial in etiology.

2.13 Pericarditis

Inflammation of the pericardial sac surrounding the heart. Presents with sharp chest pain that worsens with inspiration and lying flat (relieved by sitting forward), fever, tachycardia, and a characteristic pericardial friction rub on auscultation. Must be distinguished from MI: in pericarditis, pain improves sitting forward and worsens lying back; MI pain is unaffected by position.

UNIT 3: RESPIRATORY PATHOPHYSIOLOGY

3.1 Obstructive Lung Diseases

Obstructive lung diseases are characterized by increased resistance to airflow, typically during expiration. Air is trapped in the lungs because the airways collapse during expiration before air can fully exit.

Chronic Bronchitis

Defined clinically as a productive cough for at least 3 consecutive months in each of 2 consecutive years, in the absence of other explanations. The most common cause is cigarette smoking. Chronic irritation causes hypertrophy of mucus-secreting glands in the bronchial walls, producing excessive thick mucus. This mucus obstructs airflow and promotes infection. During expiration, the airways collapse under the weight of mucus and loss of elastic recoil, trapping air.

Emphysema

Emphysema involves permanent enlargement and destruction of the alveolar spaces distal to the terminal bronchioles. The primary mechanism in smokers is the release of proteases (elastase) from neutrophils and macrophages, which destroy the alveolar walls and elastic tissue. In a minority of cases, emphysema results from alpha-1-antitrypsin (AAT) deficiency — a genetic condition in which the normal protease inhibitor is absent.

Consequences: Loss of elastic recoil causes dynamic airway collapse during expiration, trapping air in the lungs. The barrel chest (increased anterior-posterior diameter) results from chronic lung hyperinflation due to this air trapping. The alveolar surface area for gas exchange is dramatically reduced.

Cyanosis: Bluish discoloration of the skin and mucous membranes resulting from inadequate oxygenation (low SpO₂). This should be documented using the term cyanosis. Hemoptysis refers to coughing up blood; hematemesis refers to vomiting blood; ischemia refers to reduced blood supply to tissue.

Asthma

Asthma is a chronic inflammatory airway disease characterized by reversible airway obstruction and airway hyperresponsiveness. The pathophysiology begins with exposure to a trigger — most commonly an allergen — which causes mast cell degranulation. Mast cells release histamine, leukotrienes, and prostaglandins, which drive bronchoconstriction, mucosal edema, and excess mucus production. IgE mediates the allergic response; asthmatic individuals typically have elevated IgE.

Airway Hyperresponsiveness: The airways of asthmatics are abnormally sensitive to stimuli that would not affect normal airways. This is related to the release of inflammatory mediators from mast cells following allergen exposure — not increased sympathetic activity (which would cause bronchodilation), not stress hormones alone, and not hereditary decreases in IgE (the opposite is true).

Status Asthmaticus: A severe, prolonged asthmatic attack that does not respond to standard bronchodilator therapy. It is a medical emergency characterized by severe bronchospasm, progressive hypoxemia, and respiratory failure. In a child with prolonged bronchospasm and severe hypoxemia, status asthmaticus is the diagnosis — not exercise-induced asthma (which is brief) or COPD (not a pediatric diagnosis).

Cystic Fibrosis

Cystic fibrosis (CF) is the most common lethal autosomal recessive genetic disease in White populations. It results from mutations in the CFTR gene, which encodes a chloride channel regulating ion and water transport across epithelial membranes. The result is abnormally thick, dehydrated mucus that obstructs the airways, pancreatic ducts, and other secretory structures.

Pulmonary Manifestations: Thick mucous sputum (characteristic of CF) obstructs the airways, providing a medium for chronic bacterial colonization (particularly *Pseudomonas aeruginosa*). Progressive lung damage leads to the most feared complication — respiratory failure — which is the leading cause of death in CF patients.

GI Manifestations: Pancreatic duct obstruction prevents digestive enzymes from reaching the intestine, causing fat malabsorption. Stools are frequent, loose, fatty (steatorrhea), and foul-smelling. Nutrient malabsorption leads to failure to thrive. Meconium ileus at birth (thick meconium obstructing the bowel) is strongly associated with CF.

Diagnosis: The sweat chloride test is the gold standard. CF patients have abnormally elevated chloride concentrations in their sweat because the defective CFTR channel cannot reabsorb chloride from sweat glands.

Inheritance

CF is autosomal recessive — both copies of the CFTR gene must be defective for the disease to manifest. Carriers (one defective copy) are unaffected. Parents of a child with CF are both obligate carriers.

Respiratory Distress Syndrome (RDS) of the Newborn

RDS occurs predominantly in premature infants because the type II pneumocytes that produce surfactant are not sufficiently mature before 36 weeks gestation. Surfactant reduces alveolar surface tension, preventing alveolar collapse at the end of expiration. Without adequate surfactant, alveoli collapse with each breath, and the infant must exert enormous effort to re-inflate them.

Most Important Risk Factor: Premature birth is the single most important predisposing factor for RDS. Prematurity is more significant than low birth weight (which often accompanies prematurity), maternal smoking, or alcohol use.

Timing of Signs: Signs of RDS appear within the first few minutes to hours of birth — not days later — because the problem is immediately evident when the infant must breathe on its own.

Atelectasis: RDS newborns are monitored for atelectasis (alveolar collapse) because the absence of surfactant causes alveoli to collapse after each expiration — they cannot remain open without the surfactant film reducing surface tension.

Bronchopulmonary Dysplasia (BPD): A chronic lung disease that develops in premature infants requiring mechanical ventilation for RDS. The prolonged oxygen therapy and ventilator-induced injury cause inflammation and fibrosis. An infant born 10 weeks premature, ventilated, and presenting 2 months later with hypoxemia and hypercapnia has BPD.

Sudden Infant Death Syndrome (SIDS)

SIDS is the sudden, unexpected death of an apparently healthy infant under 1 year of age that remains unexplained after thorough investigation. It peaks between 2 and 4 months of age. Risk factors include prone sleeping position (face down), soft sleep surfaces, overheating, male sex, preterm birth, and being Black or Native American. The critical prevention message: always place infants on their backs to sleep on a firm mattress (the 'Back to Sleep' or 'Safe to Sleep' campaign).

Croup and Epiglottitis

Croup (Laryngotracheobronchitis): A viral infection (most commonly parainfluenza virus) of the subglottic airway in young children. The hallmark is a barking, seal-like cough, stridor, and hoarseness from subglottic edema narrowing the airway. A truncal rash is NOT a feature of croup.

Epiglottitis: A bacterial infection (historically *Haemophilus influenzae* type B, now less common due to vaccination) causing rapid, life-threatening swelling of the epiglottis. Classic presentation: high fever, inspiratory stridor, drooling, dysphagia, and the child sitting in a 'tripod' position. Critical nursing action: DO NOT attempt to examine the throat — this can trigger complete airway obstruction. Prepare for immediate intubation.

Pleural Disorders

Pleural Effusion: Fluid accumulation in the pleural space. Transudative effusion contains protein-poor, watery fluid (results from hydrostatic pressure changes, as in heart failure or cirrhosis). Exudative effusion contains protein-rich fluid (results from inflammation, infection, or malignancy). The watery fluid drained from a patient with cirrhosis is transudative.

Pneumothorax: Air in the pleural space. Tension pneumothorax occurs when air enters but cannot exit the pleural space with each breath, building up under pressure exceeding atmospheric. This compresses the lungs and shifts the mediastinum (tracheal deviation) away from the affected side, impairs venous return, and rapidly causes hemodynamic compromise. Signs include hypotension, hypoxemia, absent breath sounds on the affected side, and tracheal deviation toward the contralateral side.

Pneumonia

Pneumonia is inflammation and consolidation of lung parenchyma. The most common causes are viral or bacterial. In children under 2 years, viral etiologies dominate. In adults, typical bacterial pneumonia (*S. pneumoniae*) presents with fever, productive cough, and lobar consolidation. Atypical pneumonia (*Mycoplasma pneumoniae*, *Legionella*, *Chlamydophila*) presents with a low-grade fever and dry cough in a 'walking pneumonia' pattern. Nosocomial (hospital-acquired) pneumonia is acquired during hospitalization.

Aspiration Pneumonia: Aspiration of oral secretions introduces anaerobic bacteria into the lower airways, causing pneumonia. This is an important complication in patients with impaired gag reflex, altered consciousness, or dysphagia.

Tuberculosis

Tuberculosis (TB) is caused by *Mycobacterium tuberculosis* — a slow-growing, acid-fast bacterium (not a virus, fungus, or parasite). TB is spread by airborne droplet nuclei. Primary infection is often contained by the immune system, lying dormant (latent TB). Reactivation occurs with immunosuppression. Active TB presents with productive cough, hemoptysis, night sweats, weight loss, and fever.

Orthopnea

Orthopnea is difficulty breathing when lying flat, relieved by sitting upright or using multiple pillows. It results from redistribution of interstitial fluid from the dependent extremities back into the pulmonary circulation when supine, worsening pulmonary congestion. It is a hallmark of left heart failure. To document that a patient reports difficulty breathing when lying down, the correct term is orthopnea.

UNIT 4: RENAL AND URINARY PATHOPHYSIOLOGY

4.1 Urinary Tract Infections

UTIs range from uncomplicated cystitis (bladder infection) to pyelonephritis (kidney infection). The most common causative organism is *Escherichia coli*, which colonizes the periurethral area and ascends into the bladder. Women are far more susceptible than men due to the short female urethra.

Acute Cystitis: Lower UTI characterized by frequency, urgency, dysuria (painful urination), and suprapubic discomfort. Low back pain may also occur. The diagnosis is supported by urinalysis showing bacteriuria and pyuria.

Mechanisms of Protection in Men: Men are less susceptible to UTIs due to the long male urethra (mechanical barrier), antimicrobial prostatic secretions, and the implantation of the ureters in the bladder at an oblique angle (which creates a one-way valve). Note that urine pH is typically slightly acidic — not alkaline — which is antimicrobial.

4.2 Pyelonephritis

Pyelonephritis is infection of the renal pelvis and parenchyma. It most commonly arises from ascending bacteria from the bladder (the most common predisposing condition is cystitis). The site of infection is the renal pelvis, not the glomerulus, tubules, or bladder. Clinical features include fever, flank pain, costovertebral angle tenderness, and systemic symptoms (nausea, vomiting). Imaging may reveal hydronephrosis if there is obstruction.

4.3 Glomerulonephritis

Glomerulonephritis is inflammation of the glomeruli, the filtering units of the kidney. It most commonly results from immune-mediated injury rather than direct infection of the glomeruli.

Poststreptococcal Glomerulonephritis: The classic form, occurring 1–3 weeks after a streptococcal infection of the throat (pharyngitis) or skin (impetigo). The mechanism is immune complex deposition in the glomerular capillaries — antigen-antibody complexes formed during the strep infection are deposited in the glomerular basement membrane, triggering complement activation and inflammatory damage. The infection is located in the respiratory tract (throat) in most cases.

IgA Nephropathy (Berger Disease): The most common primary glomerulonephritis worldwide. Characterized by deposition of immunoglobulin IgA in the glomerular capillaries. The hallmark clinical finding is recurrent gross hematuria, often triggered by mucosal infections (respiratory or GI). The nurse monitors for recurrent hematuria.

4.4 Nephrotic vs Nephritic Syndrome

Nephrotic Syndrome	Nephritic Syndrome
Massive proteinuria (>3.5g/day)	Hematuria (blood in urine — most common symptom)
Frothy urine (from protein)	Oliguria and hypertension
Hypoalbuminemia → edema	Proteinuria usually less severe than nephrotic
Lipiduria and hyperlipidemia	RBC casts in urine
Periorbital edema characteristic in children	Often immune-mediated (post-strep, IgA nephropathy)

Memory Trick

NEPHROTIC = protein (frothy urine, edema from hypoalbuminemia).
 NEPHRITIC = nephritis = inflammation = blood (hematuria).

4.5 Renal Failure

Prerenal: Caused by decreased perfusion to the kidneys (hypotension, dehydration, cardiac failure). The kidneys themselves are intact but receive insufficient blood flow.

Intrarenal (Intrinsic): Damage within the kidney itself — glomeruli, tubules, or interstitium. Acute tubular necrosis (ATN) is the most common cause, often from severe burns, hypotension, or nephrotoxic agents. A severely burned patient developing ATN has intrarenal failure.

Postrenal: Obstruction of urine outflow distal to the kidneys. An enlarged prostate (BPH) blocking urinary outflow causes postrenal failure. The obstruction increases pressure back up through the collecting system.

Chronic Renal Failure (CKD): Progressive, irreversible loss of nephron function. Key complications include anemia (due to inadequate erythropoietin production by failing kidneys), bone disease (from failure to activate vitamin D, which impairs calcium absorption and leads to secondary hyperparathyroidism, increasing bone resorption), and uremia. Dietary protein restriction is a cornerstone of CKD management because protein metabolism produces nitrogenous waste products (urea, creatinine) that the failing kidneys cannot adequately excrete.

4.6 Urinary Tract Obstruction and Hydronephrosis

Obstruction of urine flow at any level causes pressure to back up into the collecting system. This increases hydrostatic pressure within the renal pelvis and calyces, causing dilation (hydronephrosis). The elevated pressure is transmitted to the glomerulus, reducing the net filtration pressure and decreasing the GFR. The cause of hydronephrosis and decreased GFR is the dilation of the renal pelvis and calyces proximal to the blockage.

4.7 Renal Calculi (Kidney Stones)

The most common type of renal stone is composed of calcium (calcium oxalate or calcium phosphate), accounting for approximately 80% of cases. Struvite stones are associated with urease-producing bacterial infections. Uric acid stones occur in patients with gout or high purine intake. Cystine stones result from a genetic defect in amino acid transport.

Clinical Presentation: Renal colic — severe, colicky flank pain radiating to the groin. Hematuria is almost universally present as the stone abrades the urothelium. Flank pain is the most likely symptom of a kidney stone causing ureteral obstruction.

4.8 Incontinence Types

Type	Mechanism / Scenario
Stress incontinence	Leakage with increased intra-abdominal pressure (coughing, sneezing, laughing). Weakness of urethral sphincter or pelvic floor.
Urge incontinence	Sudden, strong urge followed by involuntary leakage. Overactive bladder/detrusor muscle.
Overflow incontinence	Bladder overfills and leaks. Caused by urethral obstruction (BPH) or underactive detrusor.

Functional incontinence	Physical or cognitive inability to reach the toilet in time despite intact urinary tract.
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4.9 Vesicoureteral Reflux (VUR)

VUR is the retrograde (backward) flow of urine from the urinary bladder into the ureters, and in severe cases into the renal pelvis and calyces. It results from an incompetent ureterovesical junction. It is a major cause of recurrent UTIs, particularly in children, and can lead to chronic pyelonephritis and renal scarring.

VUR is graded I through V based on severity: Grade I involves reflux only into the ureter; Grade V involves massive reflux into a grossly dilated, tortuous ureter with intrarenal reflux. Reflux into a grossly dilated ureter and calyces corresponds to Grade IV.

4.10 Wilms Tumor (Nephroblastoma)

Wilms tumor is the most common malignant renal tumor of childhood, arising from embryonic renal tissue. The kidneys are the site. Tumors may be bilateral. Presentation includes an abdominal mass (often found incidentally by a parent during bathing), abdominal pain, hematuria, hypertension, and fever.

4.11 Developmental Renal Anomalies

Condition	Description
Bilateral renal agenesis	Absence of both kidneys. Fatal — infants die within hours of birth due to pulmonary hypoplasia
Potter syndrome	Results from bilateral renal agenesis; classic facies: wide-set eyes, parrot-beak nose, low-set ears, receding chin; oligohydramnios causes limb deformities
Renal aplasia	Failure of the kidney to develop (rudimentary remnant only)
Renal agenesis (unilateral)	One kidney absent; remaining kidney undergoes compensatory hypertrophy — not atrophy, not renal failure
Hypoplastic kidney	Chronic renal failure from failure of development of one kidney

4.12 Enuresis

Primary enuresis: The child has never achieved consistent continence. A 7-year-old who has never been continent — primary enuresis.

Secondary enuresis: The child was previously continent and has regressed.

Nocturnal: Bedwetting at night. **Diurnal:** daytime incontinence.

UNIT 5: FEMALE REPRODUCTIVE PATHOPHYSIOLOGY

5.1 Common Gynecologic Disorders

Endometriosis

Endometriosis occurs when endometrial tissue implants and grows outside the uterus, most commonly on the ovaries, fallopian tubes, and pelvic peritoneum. This ectopic endometrial tissue responds to monthly hormonal cycles — proliferating and bleeding with each cycle, with nowhere for the blood to exit. This causes inflammation, adhesions, and the characteristic cyclic pelvic pain. Because the tissue is hormonally driven, suppressing ovulation (using hormonal contraceptives or GnRH agonists) is the most appropriate medical treatment.

Polycystic Ovary Syndrome (PCOS)

PCOS is one of the most common endocrine disorders in reproductive-age women. It is characterized by hyperandrogenism (causing hirsutism, acne, and male-pattern hair loss), oligo-ovulation or anovulation (leading to irregular or absent periods), and polycystic ovarian morphology on ultrasound. The condition is strongly associated with insulin resistance and hyperinsulinemia, which drives excess androgen production from the ovaries.

Consequences: The most clinically significant consequence is infertility due to irregular or absent ovulation. PCOS also predisposes to endometrial hyperplasia (from prolonged unopposed estrogen), metabolic syndrome, and type 2 diabetes. It does not typically cause early menopause or pelvic inflammatory disease directly.

Presentation: Amenorrhea or oligomenorrhea, hirsutism, and acne. Laboratory testing typically reveals elevated androgens and hyperinsulinemia.

Leiomyomas (Uterine Fibroids)

Leiomyomas are the most common benign tumors of the female reproductive tract. They arise from the smooth muscle cells (myometrium) of the uterine wall. They are estrogen-dependent and tend to shrink after menopause. Presentation includes abnormal uterine bleeding (heavy, prolonged menses), pelvic pressure, and pain. Ultrasound confirms benign uterine tumors in the smooth muscle cells of the myometrium.

Dysfunctional Uterine Bleeding (DUB)

DUB is abnormal uterine bleeding not attributable to structural causes. It typically results from hormonal imbalances. Contributing conditions include PCOS, obesity (excess adipose tissue converts androgens to estrogens, creating hormonal excess), thyroid disease (both hypothyroidism and hyperthyroidism disrupt the menstrual cycle), and excessive exercise (which suppresses the hypothalamic-pituitary axis). Hepatitis is not a direct contributor.

Dysmenorrhea

Dysmenorrhea means painful menstruation. Primary dysmenorrhea has no underlying pathology and results from excess prostaglandin production causing uterine cramping. Secondary dysmenorrhea is due to an underlying condition such as endometriosis or fibroids. Treatment includes NSAIDs (which inhibit prostaglandin synthesis) and hormonal contraceptives.

Ovarian Cysts and Ovarian Cancer

Dermoid cyst (teratoma): A benign ovarian cyst containing ectodermal tissue — classically contains skin, hair, teeth, cartilage, and bone. It arises from pluripotent germ cells.

Ovarian cancer staging: Stage I: cancer confined to the ovaries. Stage II: extends to the pelvis. Stage III: extends beyond the pelvis to the abdomen or lymph nodes. Stage IV: distant metastases. A CT scan revealing cancer limited to the ovaries = Stage I.

Cervical Cancer and HPV

Human papillomavirus (HPV) infection, specifically high-risk strains (types 16 and 18), is the primary etiologic factor in cervical cancer. Sexual transmission is the route of infection. HPV infection transforms cervical epithelial cells, leading to dysplasia and, if untreated, invasive carcinoma. Regular Pap smears detect pre-malignant changes. HPV vaccine provides effective prophylaxis.

Most Common Cancers of the Female Reproductive Tract

The most commonly occurring cancer of the female reproductive tract is endometrial (uterine) cancer. This is a frequently tested distinction — ovarian cancer is more lethal but less common; cervical cancer is preventable with screening but not the most common. Endometrial cancer is most common.

Breast Cancer

The most common first clinical manifestation of breast cancer is a painless lump. Painlessness is important — patients may not seek care for a lump that does not hurt. Other signs include skin dimpling (peau d'orange), nipple retraction, nipple discharge, and axillary lymphadenopathy in later stages.

Galactorrhea

Galactorrhea is breast discharge in a non-pregnant, non-lactating woman. It is most commonly caused by elevated prolactin levels (hyperprolactinemia). Prolactin is normally suppressed by dopamine from the hypothalamus. Any condition that reduces dopamine signaling — pituitary adenoma, certain medications (antipsychotics, metoclopramide) — can cause hyperprolactinemia and galactorrhea.

DES Exposure and Vaginal Cancer

Diethylstilbestrol (DES) was a synthetic estrogen given to pregnant women in the mid-20th century to prevent miscarriage. Daughters exposed in utero have an elevated risk of clear cell adenocarcinoma of the vagina — a rare but important association. If a patient's mother was given DES during pregnancy, the most likely DES-related cancer is vaginal cancer.

Pelvic Organ Prolapse

Cystocele: Descent of the bladder and anterior vaginal wall into the vaginal canal. Caused by weakening of pelvic floor support.

Rectocele: Posterior vaginal wall prolapse with the rectum bulging into the vagina.

Treatment for pelvic organ prolapse typically involves a pessary — a device inserted into the vagina to support the prolapsed structures — before surgery is considered.

Sexually Transmitted Infections — Female Aspects

Candidal (Yeast) Vaginitis: Caused by overgrowth of *Candida albicans*, a normally present fungus. When normal vaginal flora (*Lactobacillus*) is disrupted — by antibiotics, pregnancy, or immunosuppression — *Candida* overgrows. Presents with white, curdy discharge and intense pruritus.

Scabies: A parasitic STI caused by the mite *Sarcoptes scabiei*, transmitted through close skin-to-skin contact. It is a parasitic infection, not bacterial or viral.

UNIT 6: MALE REPRODUCTIVE PATHOPHYSIOLOGY

6.1 Benign Prostatic Hyperplasia (BPH)

BPH is a non-malignant enlargement of the prostate gland that occurs with aging. The most important risk factor is increased age. As the prostate enlarges, it compresses the urethra, creating bladder outlet obstruction. The classic complaint is a decreased urinary stream (weak, slow flow), along with frequency, nocturia, hesitancy, and incomplete bladder emptying. If severe, BPH can cause postrenal failure.

6.2 Prostatitis

Acute bacterial prostatitis produces symptoms very similar to a urinary tract infection: dysuria, urinary frequency, fever, and perineal pain. This clinical overlap exists because the infecting organisms are the same (*E. coli*) and the inflammatory process affects the structures through which urine flows. The key distinction from BPH is the acute infectious component.

6.3 Prostate Cancer

Prostate cancer is the leading type of non-skin cancer in men in the United States. It is primarily a disease of older men (uncommon before 50). It is more common in Black men than White men in the United States — this is an important epidemiological fact. Death rates from prostate cancer have been declining in the United States due to improved screening (PSA testing) and treatment. It is NOT responsible for the majority of cancer-related deaths in males — that distinction belongs to lung cancer.

6.4 Cryptorchidism

Cryptorchidism refers to failure of one or both testes to descend from the abdomen into the scrotum by birth. It is a disorder of testicular descent, not testicular overdevelopment or lack of scrotum.

Cryptorchidism is a significant risk factor for testicular cancer, likely due to elevated temperature in the undescended testis impairing normal spermatogenesis and potentially initiating malignant transformation. Surgical correction (orchiopexy) is performed early to reduce this risk.

6.5 Testicular Cancer

Testicular cancer predominantly affects young men between 15 and 35 years of age and those of White/European ethnicity. Risk is elevated with a history of cryptorchidism. It is one of the most curable cancers when detected early.

6.6 Orchitis

Orchitis is inflammation of the testes. The most common cause in post-pubertal males is mumps virus infection (paramyxovirus), which can spread hematogenously to the testes and cause painful swelling. If bilateral, orchitis from mumps can cause sterility.

6.7 Penile Disorders

Condition	Description
Phimosis	Foreskin cannot be retracted back over the glans penis (too tight). May cause difficulty urinating.

Paraphimosis	Retracted foreskin becomes trapped behind the glans and cannot be pulled forward — a urological emergency as it can compromise circulation.
Peyronie disease	Fibrous plaques form in the tunica albuginea, causing penile curvature during erection and pain with intercourse.
Balanitis	Inflammation of the glans penis. Often associated with poor hygiene or phimosis.
Priapism	Prolonged, painful erection unrelated to sexual stimulation — a medical emergency.
Hydrocele	Collection of fluid around the testis — not a penile disorder.

6.8 Common STIs — Male Presentation

Chlamydia (*Chlamydia trachomatis*): The most common bacterial STI in the United States. In men, it commonly causes urethritis and epididymitis. It is often asymptomatic, contributing to its widespread transmission.

Gonorrhea (*Neisseria gonorrhoeae*): The second most common bacterial STI. Causes purulent urethral discharge and dysuria in men.

Syphilis (*Treponema pallidum*): Caused by the spirochete *T. pallidum*. Progresses through primary (painless chancre), secondary (diffuse rash including palms and soles), and tertiary (cardiovascular and neurological) stages.

6.9 Gynecomastia

Gynecomastia is the overdevelopment (enlargement) of breast tissue in males. It results from an imbalance between estrogens and androgens. Certain medications cause gynecomastia as a side effect, including spironolactone, cimetidine, anabolic steroids, and some antipsychotics. It should not be confused with pseudogynecomastia (adipose tissue), breast cancer, or mastitis.

6.10 Delayed Puberty

In the majority of cases, delayed puberty is due to constitutional delay — simply slow maturation — and is not pathological. The hypothalamic-pituitary-gonadal axis is intact but delayed. True disruptions of the hypothalamus or pituitary gland are less common causes. Constitutional delay is most common.

UNIT 7: GASTROINTESTINAL AND HEPATIC PATHOPHYSIOLOGY

7.1 Esophageal and Gastric Disorders

GERD and Hiatal Hernia

Gastroesophageal reflux disease (GERD) occurs when gastric contents reflux into the esophagus. The primary cause is loss of muscle tone (incompetence) at the lower esophageal sphincter (LES), which normally creates a pressure barrier preventing reflux. The refluxed acid damages the esophageal mucosa. Complications include esophagitis, Barrett esophagus (metaplastic columnar epithelium replacing squamous — a pre-malignant change), esophageal stricture, chronic cough, and in children, recurrent otitis media.

A hiatal hernia occurs when part of the stomach herniates through the esophageal hiatus of the diaphragm into the thoracic cavity. Sliding hiatal hernia is most common and predisposes to GERD. Paraesophageal hernia is less common but more dangerous — the fundus herniates beside the esophagus, risking strangulation (compromise of blood supply) as the most serious complication.

Peptic Ulcer Disease (PUD)

Peptic ulcers are breaks in the mucosal lining of the stomach or duodenum where acid and pepsin cause tissue damage. The two leading causes are *Helicobacter pylori* infection (a gram-negative bacterium that disrupts the mucus layer) and chronic NSAID use (which inhibits prostaglandin synthesis, reducing the protective mucus layer). Regular NSAID use contributes to ulcer development.

Gastric Resection Complications: After gastrectomy, patients lose the stomach's role in acid production and intrinsic factor secretion. Loss of intrinsic factor leads to anemia (B12 deficiency). Dumping syndrome is another complication. Of the listed choices, anemia is the complication most directly linked to gastric resection.

GI Bleeding Terminology

Melena: Black, tarry stool resulting from digested blood — indicates upper GI bleeding.

Hematochezia: Bright red blood per rectum — indicates lower GI bleeding or very brisk upper GI hemorrhage.

Hematemesis: Vomiting blood — from upper GI source.

Occult bleeding: Blood in stool not visible to the naked eye, detected by chemical testing.

Visceral vs Somatic Abdominal Pain

Visceral pain: From stimulation of receptors in the visceral peritoneum covering abdominal organs. Poorly localized, crampy, midline. Midline epigastric pain from abdominal organ stimulus is visceral pain.

Somatic/parietal pain: From stimulation of the parietal peritoneum. Well-localized, sharp, exacerbated by movement (rebound tenderness).

Referred pain: Felt at a site distant from the organ of origin (e.g., shoulder pain from diaphragm irritation).

Appendicitis

Appendicitis is inflammation of the vermiform appendix, typically due to obstruction followed by bacterial infection. Classic presentation: periumbilical pain migrating to the right lower quadrant (McBurney's point), rebound tenderness, and low-grade fever. It is a surgical emergency.

Diarrhea Types

Type	Mechanism and Example
Osmotic	Unabsorbed solutes draw water into the bowel lumen. Example: lactose intolerance — undigested lactose creates an osmotic load.
Secretory	Enterotoxins stimulate active secretion of fluid/electrolytes. Example: bacterial GI infection (cholera, E. coli enterotoxin). This is the most likely type in bacterial infections.
Motility	Abnormally rapid or slow transit. Example: IBS.

Mesenteric Ischemia

Acute mesenteric arterial insufficiency: Sudden occlusion of mesenteric vessels causes intestinal infarction. Ischemia alters mucosal membrane permeability, causing massive fluid loss into the bowel lumen and peritoneum — this is why hypovolemic shock risk is high, not from direct GI bleeding.

Chronic mesenteric ischemia: The most common cause among the elderly is atherosclerosis of the mesenteric arteries, leading to postprandial abdominal pain ('intestinal angina').

7.2 Hepatic and Pancreatic Disorders

Hepatitis

Hepatitis is inflammation of the liver, most commonly from viral infection. The different hepatitis viruses have distinct transmission routes and incubation periods.

Type	Transmission	Key Features
Hepatitis A	Fecal-oral; poor handwashing	Outbreaks in day care settings; acute, self-limiting; no chronic form
Hepatitis B	Blood, sexual, perinatal	Incubation up to 180 days; can become chronic; Coombs negative
Hepatitis C	Blood (IV drug use, transfusions)	Incubation up to 180 days; most common cause of chronic liver disease; high risk of chronicity
Hepatitis D	Blood; only with HBV co-infection	Requires HBV to replicate; incubation up to 180 days

Hepatitis E	Fecal-oral	Similar to HAV; dangerous in pregnancy
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Phases of Hepatitis: The icteric phase (jaundice phase) is characterized by jaundice, dark urine (from elevated conjugated bilirubin), pale/clay-colored stools, and an enlarged liver. This is distinct from the prodromal phase (fatigue, malaise, vomiting) and the recovery phase (resolution of jaundice, liver function returns to normal).

Jaundice Types

Type	Mechanism
Hemolytic (pre-hepatic)	Increased rate of RBC breakdown produces more bilirubin than the liver can process. Unconjugated bilirubin rises.
Hepatocellular	Damaged liver cells cannot conjugate and excrete bilirubin normally.
Obstructive (post-hepatic)	Bile duct obstruction prevents bile excretion. Conjugated bilirubin backs up into blood.

In a newborn with a lack of bilirubin uptake by hepatocytes (physiologic immaturity), the result is physiologic jaundice — a normal transient condition in the first week of life.

Cirrhosis and Portal Hypertension

Cirrhosis is end-stage liver disease characterized by replacement of normal hepatic tissue with fibrous scar tissue and regenerative nodules. The most common cause of cirrhosis with the classic triad of enlarged liver, testicular atrophy, and mild jaundice is alcoholic steatohepatitis (alcoholic liver disease).

Portal hypertension results from increased resistance to blood flow through the fibrotic liver. Blood cannot flow easily through the portal system, causing pressure to back up. The most common cause of portal hypertension is cirrhosis. Consequences include esophageal varices (which can rupture, causing hematemesis), ascites, and splenomegaly.

Hepatic Encephalopathy: A neuropsychiatric complication of chronic liver disease. The failing liver cannot metabolize ammonia (a byproduct of protein catabolism and bacterial action in the gut). Ammonia accumulates and crosses the blood-brain barrier, causing cognitive impairment, asterixis (flapping tremor), and coma. The mechanism is impaired ammonia metabolism, not hyperbilirubinemia or fluid imbalances.

Extrahepatic Portal Hypertension: When the obstruction is in the hepatic portal vein itself (rather than within the sinusoids), portal hypertension is termed extrahepatic. The nurse would expect the obstruction in the hepatic portal vein.

Pancreatitis

Acute pancreatitis results from premature activation of digestive enzymes within the pancreas itself — a backup of pancreatic enzymes triggers autodigestion. The pain is characteristically epigastric, often radiating to the back, and intensifies after eating. The two most common causes are gallstones (blocking the pancreatic duct) and alcohol use.

Treatment of acute pancreatitis focuses on bowel rest (restriction of oral food intake to reduce pancreatic stimulation), nasogastric suctioning in severe cases, IV fluid replacement, and narcotic analgesics for pain. Steroids are not standard treatment.

Chronic pancreatitis contributing factors include alcohol abuse, history of gallstones, trauma, and smoking. Bulimia is not a direct contributor.

Cholecystitis and Gallstones (Cholelithiasis)

Cholecystitis is inflammation of the gallbladder wall, most commonly caused by obstruction of the cystic duct by a gallstone, leading to bile stasis and inflammation. The majority of gallstones have a high concentration of cholesterol, resulting from supersaturated bile. Stones can also be pigmented (bilirubin-based), but cholesterol stones are most common.

7.3 Pediatric GI Disorders

Pyloric Stenosis

Pyloric stenosis is hypertrophy of the pyloric sphincter muscles, causing progressive obstruction of gastric outflow. It typically presents in infants aged 2–8 weeks. The classic signs are projectile (forceful, non-bilious) vomiting immediately after feeding, constipation, and a palpable olive-sized mass in the upper abdomen (the hypertrophied pylorus). The vomiting is non-bilious because the obstruction is above the point where bile enters (the duodenum).

Esophageal Atresia

Esophageal atresia occurs when the esophagus ends in a blind pouch rather than connecting to the stomach. Often associated with a tracheoesophageal fistula. The infant cannot swallow amniotic fluid, leading to excessive amniotic fluid accumulation during pregnancy (polyhydramnios). Nurses should assess for polyhydramnios in mothers carrying fetuses with esophageal malformations. After birth, feeding attempts result in choking and aspiration.

Intussusception

Intussusception occurs when one portion of the bowel telescopes into an adjacent segment, causing obstruction. The classic presentation is in infants 6–18 months: sudden, severe, intermittent abdominal pain, vomiting, and passage of currant jelly stool (a mixture of mucus and blood from mucosal ischemia). The ileum collapsing through the ileocecal valve is the most common site.

Hirschsprung Disease (Congenital Megacolon)

Hirschsprung disease results from failure of neural crest cells to migrate into the distal colon during fetal development, leaving a segment of bowel without intrinsic ganglion cells (the myenteric and submucosal plexuses). Without these nerve cells, the affected bowel cannot undergo peristalsis, remaining in tonic contraction. This creates a functional obstruction. Stool accumulates proximal to the aganglionic segment, causing massive dilation (megacolon). The primary complication is enterocolitis from fecal impaction.

Necrotizing Enterocolitis (NEC)

NEC is an inflammatory bowel necrosis predominantly affecting premature neonates. Presentation: abdominal distention, bilious vomiting, bloody stools, and systemic signs including unstable temperature, bradycardia, and apnea. Radiographic findings: portal venous gas and pneumatosis intestinalis (gas in the bowel wall). Intestinal perforation is a life-threatening complication.

Malrotation of the Intestine

Failure of the intestine to rotate properly during fetal development. Bile-stained (bilious) vomiting in a young infant is a red flag for malrotation with midgut volvulus. X-ray reveals a paucity of gas or characteristic 'double bubble' sign. Bilious vomiting in a 2-month-old — think malrotation.

Biliary Atresia

Biliary atresia is the absence or obstruction of intrahepatic or extrahepatic bile ducts. Without bile drainage, bilirubin accumulates. The primary clinical manifestation is jaundice persisting beyond the first few weeks of life, along with clay-colored (acholic) stools, dark urine, and an enlarged liver. If untreated, it progresses to cirrhosis.

Meckel Diverticulum

A remnant of the omphalomesenteric duct — the most common congenital anomaly of the GI tract. The diverticulum may contain ectopic gastric or pancreatic tissue. The most common symptom in children is painless rectal bleeding, from acid secretion by ectopic gastric mucosa ulcerating the adjacent bowel wall.

Cleft Lip

Cleft lip results from incomplete fusion of the nasomedial process during early fetal development, typically visible on ultrasound by the late first trimester. It is a common congenital facial anomaly that can occur alone or with cleft palate.

Down Syndrome and GI Complications

Infants with Down syndrome (trisomy 21) and congenital heart disease have an elevated risk of imperforate anus — a congenital malformation in which the anal opening is absent or abnormally positioned. This is a surgically correctable condition.

UNIT 8: MUSCULOSKELETAL PATHOPHYSIOLOGY

8.1 Fracture Types and Healing

A fracture is a break in the continuity of bone. Understanding fracture terminology is essential for documentation and clinical communication.

Fracture Type	Description
Transverse	Fracture line runs straight ACROSS (perpendicular to the long axis) the bone shaft. Do not confuse with 'parallel to the bone.'
Oblique	Fracture line at an angle to the bone shaft
Spiral	Fracture winds around the shaft; from torsional forces
Comminuted	Bone is shattered into multiple fragments
Greenstick	Incomplete fracture (like bending a green twig); occurs in children whose bones are more flexible
Fatigue (stress)	Results from repeated stress on bone, overwhelming its remodeling capacity. Classic scenario: an obese person beginning a new jogging routine. Also called a stress fracture.
Compound (open)	Bone pierces the skin; risk of infection
Transchondral	Fracture through articular cartilage; most prevalent in adolescents

Dislocation vs Subluxation: Dislocation = complete loss of contact between articular surfaces. Subluxation = PARTIAL loss of contact. A broken clavicle with bone surfaces partially losing contact = subluxation.

Sprain vs Strain: Sprain = torn ligament (bone-to-bone connection). Strain = injured muscle or tendon.

Pelvic Fractures: Most common in older adults, whose bones are weakened by osteoporosis.

Transchondral Fractures: Most prevalent in adolescents whose articular cartilage is still developing.

8.2 Bone Density Disorders

Osteoporosis

Osteoporosis is characterized by decreased bone density, resulting from an imbalance between bone formation (osteoblasts) and bone resorption (osteoclasts). In osteoporosis, bone resorption exceeds formation — too much bone is broken down relative to what is built. The remaining bone is structurally intact but insufficient in quantity, increasing fracture risk. Hip fractures in elderly women are a classic complication.

Osteomalacia and Rickets

Osteomalacia is inadequate bone mineralization in adults — the organic bone matrix (osteoid) is laid down normally but is not properly calcified due to vitamin D deficiency (most commonly). Vitamin D is essential for calcium absorption from the gut. Without it, calcium cannot be adequately absorbed, and newly formed bone matrix remains unmineralized and weak. Bariatric surgery impairs vitamin D absorption — hence the link to osteomalacia.

Rickets is the equivalent condition in growing children — defective mineralization of the growth plate and newly formed bone. The bones are soft, causing bowing of the legs and other skeletal deformities.

Paget Disease

Paget disease of bone is a disorder of accelerated but disorganized bone remodeling. Osteoclasts first excessively resorb bone, followed by compensatory osteoblast activity producing structurally weak, enlarged, mosaic bone. The cause is currently unknown (believed to involve viral and genetic factors, but no definitive cause has been established). Affected bones are prone to fracture and deformity.

8.3 Osteomyelitis

Osteomyelitis is bone infection. The primary causative organism is *Staphylococcus aureus* in the vast majority of cases — regardless of patient age or route of infection. Infection reaches bone either hematogenously (via the bloodstream), by direct extension from adjacent soft tissue infection, or through open fractures. *Salmonella* is associated with osteomyelitis in patients with sickle cell disease specifically.

8.4 Arthritis

Rheumatoid Arthritis (RA)

RA is a systemic autoimmune disease characterized by chronic inflammation of synovial membranes (the lining of joints). This is the primary site of inflammation — not subchondral bone or bursae directly. The immune system produces antibodies (rheumatoid factor, anti-CCP) that attack the synovium, causing progressive joint destruction. RA affects small joints symmetrically (hands, wrists, feet), distinguishing it from OA. Extra-articular manifestations include rheumatoid nodules, vasculitis, and pulmonary involvement.

Treatment: Disease-modifying antirheumatic drugs (DMARDs) are the treatment of choice, with methotrexate being the anchor drug. NSAIDs are used for symptom control but do not alter disease progression. Gold salts were historical treatments. Colchicine is used for gout.

Osteoarthritis (OA)

OA is a degenerative joint disease characterized by progressive loss of articular cartilage — the smooth cartilage covering joint surfaces. As cartilage erodes, bone-on-bone contact causes pain, stiffness, and decreased range of motion. OA typically affects weight-bearing joints (knees, hips) and the hands. Unlike RA, OA is not primarily an inflammatory or autoimmune condition, though inflammation plays a secondary role.

Gouty Arthritis

Gout results from hyperuricemia — elevated blood uric acid — leading to deposition of monosodium urate crystals in the synovial fluid and periarticular tissues. These crystals trigger an acute inflammatory response causing the classic presentation of sudden-onset, exquisitely painful joint inflammation (most often the first metatarsophalangeal joint — the big toe). The inflammatory symptoms are caused by crystallization of monosodium urate within synovial fluid.

Risk Factors: Men aged 40–50 are at highest risk. Premenopausal women are largely protected by estrogen's uricosuric effect. Complication: uric acid renal stones are a significant risk in gout patients (gout and renal stones share the purine metabolism pathway).

Purines vs Uric Acid: Purines are the precursors to uric acid — the crystals themselves are monosodium urate, not purines or pyrimidines.

Ankylosing Spondylitis (AS)

AS is a chronic inflammatory arthritis that primarily affects the axial skeleton. The most likely joint to be affected first is the sacroiliac joint — inflammation here is pathognomonic of early AS. Progressive spinal fusion leads to a rigid spine ('bamboo spine' on X-ray). AS is strongly associated with the HLA-B27 antigen.

8.5 Bone Tumors

Secondary chondrosarcoma: A malignant tumor arising from cartilage that develops from a preexisting benign lesion (such as an osteochondroma). CT revealing a tumor arising from a preexisting benign bone lesion in the knee area = secondary chondrosarcoma.

Malignant bone tumors — patterns of bone destruction: Permeative and moth-eaten patterns indicate aggressive malignancy. Geographic pattern indicates slower-growing, more defined lesions.

8.6 Muscle Disorders

Disuse atrophy: Decrease in muscle fiber size and strength from lack of use. A patient confined to bed rest for 3 weeks experiences disuse atrophy.

Rhabdomyolysis: Rapid breakdown of skeletal muscle releasing myoglobin into the bloodstream, which can cause acute renal tubular injury. Diagnosed by measuring creatine kinase (CK), which is released from damaged muscle cells and markedly elevated.

Toxic myopathy: The most common cause is alcoholism. Alcohol directly damages muscle cells.

Epicondylopathy (tennis elbow): Inflammation of the tendon at its attachment to bone (enthesopathy), with microtears at the tendon-bone junction. MRI reveals inflammation and microtears where the tendon attaches to bone — this is epicondylopathy, not periostitis (bone surface inflammation), bursitis (bursal inflammation), or muscle strain.

UNIT 9: INTEGUMENTARY SYSTEM AND BURNS

9.1 Skin Structure and Function

The skin is the body's largest organ, serving as a barrier against infection, regulating temperature, and providing sensory input. Key structural components include the epidermis (outer layer, primarily keratinocytes), dermis (connective tissue layer with blood vessels, nerves, and dermal appendages), and hypodermis (subcutaneous adipose).

Dermal appendages: Include apocrine glands, eccrine (sweat) glands, nails, hair, and sebaceous glands. Dermal papillae are structural projections of the dermis into the epidermis — they are not appendages.

Langerhans cells: Antigen-presenting dendritic cells in the epidermis that initiate immune responses. They are the skin's immune sentinels.

Merkel cells: Mechanoreceptors for light touch.

Melanocytes: Produce melanin pigment in the epidermis.

9.2 Burns

Classification by Depth

Classification	Depth of Injury	Appearance / Features
First Degree (Superficial)	Epidermis only	Red, painful, dry; no blisters; like a sunburn
Superficial Partial-Thickness	Epidermis + superficial dermis	Blistering, very painful, moist, blanches with pressure
Deep Partial-Thickness	Epidermis + deep dermis; FEW appendages intact	Waxy white, decreased pain (nerve damage), slow to heal
Third Degree (Full-Thickness)	All layers of skin; no viable appendages	Leathery, eschar, painless (nerve destruction); needs grafting

Clinical Key	Deep partial-thickness burn: burned through ALL the dermis with only a FEW epidermal appendages intact. The burn appears waxy white. Third-degree burns destroy ALL dermal appendages completely.
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Rule of Nines / Lund and Browder

The rule of nines and the Lund and Browder chart are tools used to estimate total body surface area (TBSA) burned — not the depth, possibility of infection, or systemic involvement. TBSA determines fluid resuscitation requirements (Parkland formula).

Burn Pathophysiology

First 24 hours: Massive capillary permeability increase allows plasma proteins and fluid to shift from the vascular space into the interstitium and burn wound. This causes hypovolemia, hemoconcentration (elevated hematocrit — because water leaves but RBCs stay), and tachycardia with hypotension.

Lab Finding

High hematocrit in a burn patient = fluid movement OUT of the vascular space (hemoconcentration), not sickle cell syndrome or renal failure. Water leaves but red blood cells remain, concentrating the blood.

9.3 Pressure Ulcers

Pressure ulcers develop from sustained pressure over bony prominences, causing ischemia and tissue necrosis. The earliest sign of a developing pressure ulcer is redness (non-blanchable erythema over the bony prominence). Effective wound healing in superficial ulcers requires a flat, moist dressing — moisture maintains a healing environment for keratinocyte migration. Dry dressings impede healing.

9.4 Common Skin Infections and Infestations

Herpes Zoster (Shingles): Reactivation of the varicella-zoster virus (chickenpox) that has remained dormant in dorsal root ganglia. Occurs when cell-mediated immunity wanes (due to aging, immunosuppression). A transplant patient on immunosuppressives with a painful vesicular rash on the face and trunk, who had chickenpox as a child, has herpes zoster.

Candidiasis: Fungal infection by *Candida albicans*. Affects warm, moist areas — most commonly mucous membranes and the vagina. Frequent candidiasis suggests impaired immunity. The plantar surface of the foot is not a characteristic site.

Kaposi Sarcoma (KS): A vascular malignancy associated with human herpesvirus 8 (HHV-8). Occurs almost exclusively in immunosuppressed individuals — HIV patients and organ transplant recipients on immunosuppressive therapy. Immunosuppression is the key link.

Pityriasis Rosea: A self-limiting skin condition characterized by a herald patch followed by a generalized rash. Caused by a virus (likely HHV-6 or HHV-7).

9.5 Inflammatory Skin Conditions

Discoid Lupus Erythematosus: A chronic autoimmune skin condition causing raised red lesions with a brownish scale, primarily on sun-exposed areas. The cause is autoimmunity — not infection, trauma, or cancer.

Systemic Scleroderma: An autoimmune connective tissue disease causing widespread fibrosis. Raynaud phenomenon is a characteristic early feature, occurring in the majority of patients with scleroderma.

Pemphigus: A chronic, autoimmune blister-forming disease affecting the skin and oral mucous membranes. Autoantibodies target desmoglein proteins in desmosomes, causing loss of epidermal cell adhesion (acantholysis) and blister formation within the epidermis.

Psoriasis: A chronic immune-mediated disease causing rapid keratinocyte proliferation, resulting in thickened, silvery-scaled plaques on the skin.

9.6 Skin Lesions and Pigmentation

Nevi (moles): Benign melanocytic lesions. They are monitored because of their potential to transform into malignant melanoma. An adolescent with concerning pigmented lesions should be monitored for malignant transformation — these lesions are called nevi.

Keloids: Abnormal, exuberant scars that extend beyond the original wound boundaries due to excessive collagen deposition. Individuals of Black African descent are significantly more prone to keloid formation.

9.7 Hair and Pruritus

Alopecia: Hair loss in general. Alopecia areata refers to patchy hair loss from an autoimmune attack on hair follicles. Male pattern baldness (androgenetic alopecia) — frontotemporal hair recession — is a form of alopecia.

Hirsutism: Excessive, male-pattern hair growth in women, on the face and body. Caused by androgen excess (PCOS, adrenal disorders). The correct medical term for abnormal hair growth on the face and body in a female is hirsutism.

Pruritus: The sensation of itch. Substance P is the primary neurotransmitter associated with the itch sensation — increased substance P activity is associated with pruritus.

Quick-Reference Summary Tables

Use the following tables as a rapid review tool before assessments.

Key 'Most Common' Facts

Condition	Most Common / Key Fact
Most common childhood leukemia	ALL (Acute Lymphoblastic Leukemia)
Most common type of renal stone	Calcium oxalate
Most common bacterial STI in USA	Chlamydia
Most common cause of osteomyelitis	Staphylococcus aureus
Most common congenital heart defect	VSD (Ventricular Septal Defect)
Most common cardiac valve disease in USA	Mitral valve prolapse
Most common cancer of female reproductive tract	Endometrial cancer
First sign of breast cancer	Painless lump
Most common GI congenital anomaly	Meckel diverticulum
Most common UTI organism	Escherichia coli
Most common cause of pyelonephritis	Cystitis (ascending infection)
Most common gallstone composition	Cholesterol
Most common form of delayed puberty	Constitutional delay (slow maturation)
Most common cause of chronic bronchitis	Cigarette smoke
Most common cause of emphysema in young patients	Alpha-1-antitrypsin deficiency

Inheritance Patterns

Condition	Inheritance
Cystic fibrosis	Autosomal recessive
Thalassemia (alpha and beta)	Autosomal recessive
Sickle cell disease	Autosomal recessive
Hemophilia A and B	X-linked recessive (males affected)
Polycythemia vera	Acquired somatic mutation (JAK2)

Anemia: Morphology and Mechanism

Morphology	Mechanism	Condition
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Microcytic, hypochromic	Decreased hemoglobin synthesis from iron deficit	Iron deficiency anemia
Macrocytic, normochromic	Defective DNA synthesis (impaired cell division)	B12 or folate deficiency / pernicious anemia
Normocytic, normochromic	Premature RBC destruction	Hemolytic anemia
Mixed picture	Ineffective iron incorporation into hemoglobin	Sideroblastic anemia
Microcytic	Defective hemoglobin chain synthesis	Thalassemia

"Understanding the WHY makes you a better clinician than memorizing the WHAT."

— Dr. Vidya, SCI 225 Pathophysiology, Nightingale College